



Text-to-Speech (TTS) & Speech-to-Text (STT)

2411CS020571 – Surya

2411CS020624 – Rohith

2411CS020548 - Rudraksh

Introduction to Voice AI

Voice AI refers to technologies that enable computers to understand, process, and generate human speech. These systems bridge the gap between human communication and machine interaction, making technology more natural and accessible.

Why It Matters

Speech technologies power everyday tools — from virtual assistants to live captions — transforming how we interact with devices.

Everyday Applications

Siri, Alexa, Google Assistant, YouTube auto-captions, GPS navigation, and voice banking all rely on TTS and STT.

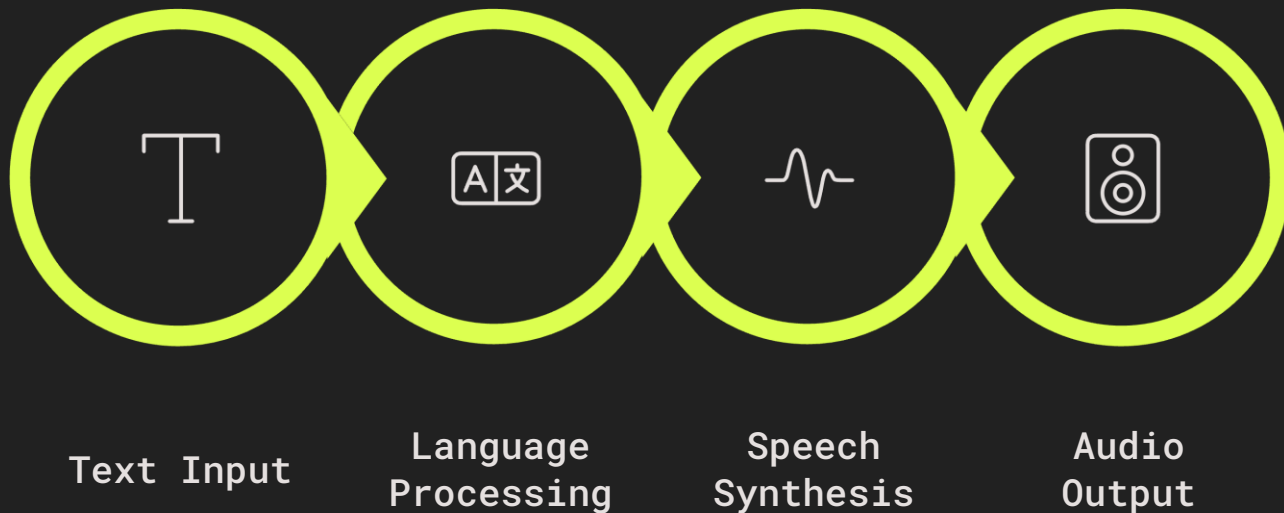
How It Works

Computers process speech using AI, machine learning, and NLP to convert between audio signals and text.



What is Text-to-Speech (TTS)?

Text-to-Speech (TTS) is a technology that converts written text into natural, human-like speech. It enables devices to "read aloud" any digital text, making information accessible through audio output. TTS powers virtual assistants, GPS navigation, audiobooks, accessibility tools for visually impaired users, customer support systems, and education platforms.



This pipeline transforms raw text into intelligible, natural-sounding audio through a series of AI-driven stages.



Components of TTS



Text Analysis

Parses raw text, identifies words, punctuation, and sentence structure.



Linguistic Processing

Applies grammar rules, resolves abbreviations, and determines pronunciation.



Phoneme Generation

Converts processed text into phonetic units representing speech sounds.



Speech Synthesis & Audio Output

Generates natural voice audio from phonemes using neural network models.

TTS: Advantages & Limitations

□ Advantages

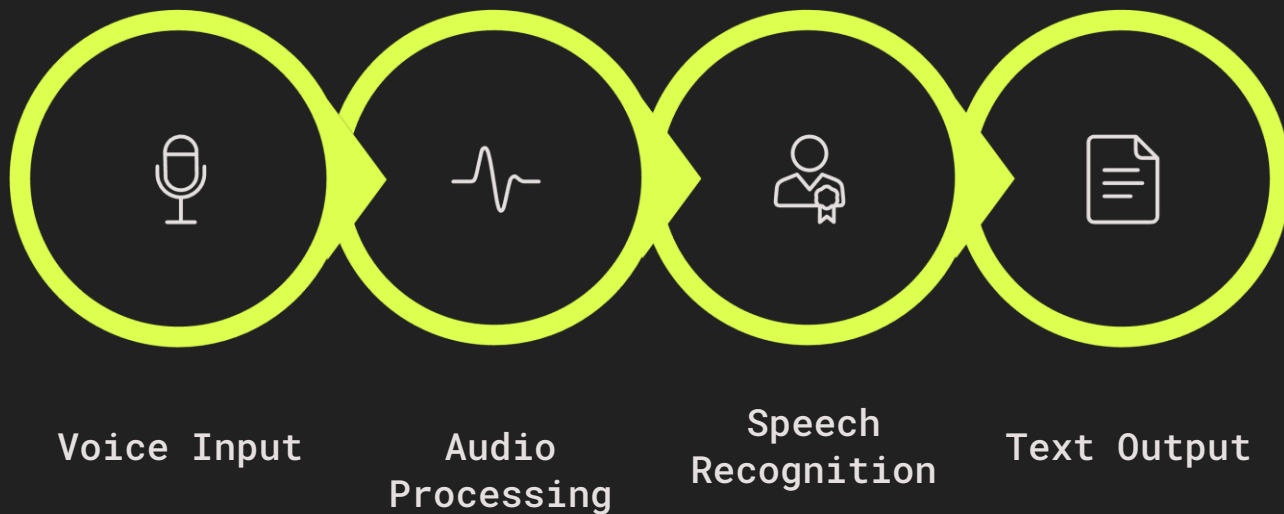
- Improves accessibility for visually impaired users
- Saves time by enabling audio consumption of text
- Supports multiple languages and dialects
- Modern neural TTS produces highly natural voices

□ □ Limitations

- Pronunciation challenges with unusual words or names
- Emotional expression and intonation remain limited
- High-quality voices require powerful AI models and computing resources

What is Speech-to-Text (STT)?

Speech-to-Text (STT) converts spoken language into written text in real time or from recorded audio. It enables hands-free interaction with devices and automates transcription tasks across industries. STT applications include voice typing, live captioning, voice search, medical documentation, call centers, smart assistants, and meeting transcription.



Each stage refines the audio signal until it becomes accurate, readable text.

Components of STT

1 Audio Capture
Microphone records spoken input as a raw audio signal.

2 Noise Reduction
Filters background noise to isolate the speaker's voice.

3 Feature Extraction
Identifies key acoustic features like frequency and pitch.

4 Acoustic Model
Maps audio features to phonetic units using trained AI.

5 Language Model
Predicts word sequences to improve transcription accuracy.

6 Text Generation
Outputs the final transcribed text to the user or application.

STT: Advantages & Limitations

□ Advantages

- Faster than manual typing for many users
- Enables hands-free operation in various contexts
- Improves productivity in documentation and note-taking
- Enhances accessibility for users with motor or visual impairments

□ □ Limitations

- Sensitive to background noise and poor audio quality
- Accent and dialect recognition remains challenging
- Privacy concerns with voice data collection
- Cloud-based services require internet connectivity

TTS vs. STT: Side-by-Side Comparison

Aspect	Text-to-Speech (TTS)	Speech-to-Text (STT)
Input	Written text	Spoken audio
Output	Audio / Speech	Written text
Purpose	Make text audible	Make speech readable
Technologies	NLP, Speech Synthesis, Neural Networks	Acoustic Models, Language Models, Deep Learning
Applications	Virtual assistants, audiobooks, accessibility, GPS	Voice typing, live captions, transcription, voice search
Examples	Google TTS, Amazon Polly, ElevenLabs	OpenAI Whisper, Google Speech, Azure STT

AI Technologies, Tools & The Future of Voice AI

TTS and STT rely on **AI, Machine Learning, Deep Learning, NLP, Neural Networks**, and **Large Language Models**. Leading platforms include Google Cloud Speech/TTS, Microsoft Azure Speech, Amazon Polly/Transcribe, OpenAI Whisper, and ElevenLabs — powering real-world examples like Siri, Alexa, Google Assistant, ChatGPT Voice, YouTube Auto Captions, and Voice Banking.

Emotion-Aware Speech

AI voices that convey tone, mood, and emotional context.

Real-Time Translation

Instant multilingual speech conversion breaking language barriers.

Personalized Voices

Custom AI voice cloning for individuals and brands.

Healthcare & Robotics

Voice AI in medical documentation, smart robotics, and assistive care.

Conclusion: TTS and STT are foundational to modern AI, enabling natural human-machine interaction. As these technologies advance, they will unlock new opportunities in accessibility, productivity, and global communication.

📖 **References:** [Google Cloud Documentation](#) · [Microsoft Azure Speech Documentation](#) · [OpenAI Whisper Documentation](#) · [Amazon Polly Documentation](#) · [IEEE Papers on Speech Processing](#)

Screenshots of Output



Sign In

Sign Up

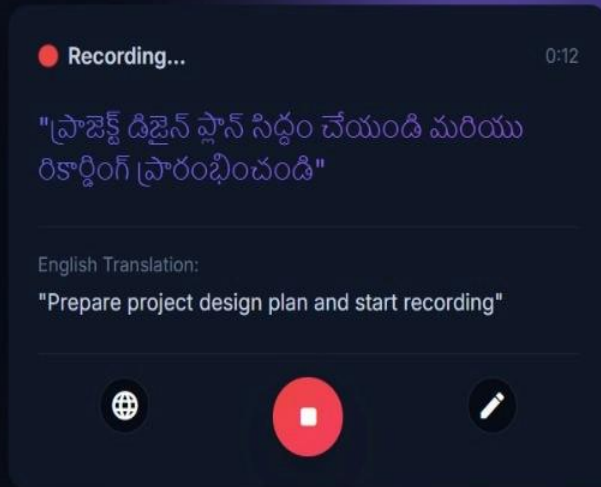
New: Multi-language Speech API

Capture Your Thoughts With Your Voice

Create rich, searchable notes by speaking. Support for English, Hindi, and Telugu transcribing, plus instant text translations. Organize, pin, favorite, and archive effortlessly.

Get Started Free

Explore API



Screenshots of Output



Welcome Back

Please enter your credentials to log in

Email or Phone Number

name@example.com or +1234567890

Password

[Forgot Password?](#)

Sign In →

Don't have an account? [Sign Up](#)

Demo Credentials:

Email: john@example.com

Password: password123

Screenshots of Output



Security Verification

We have sent a verification code to:
kyanamrudrakshrao@gmail.com

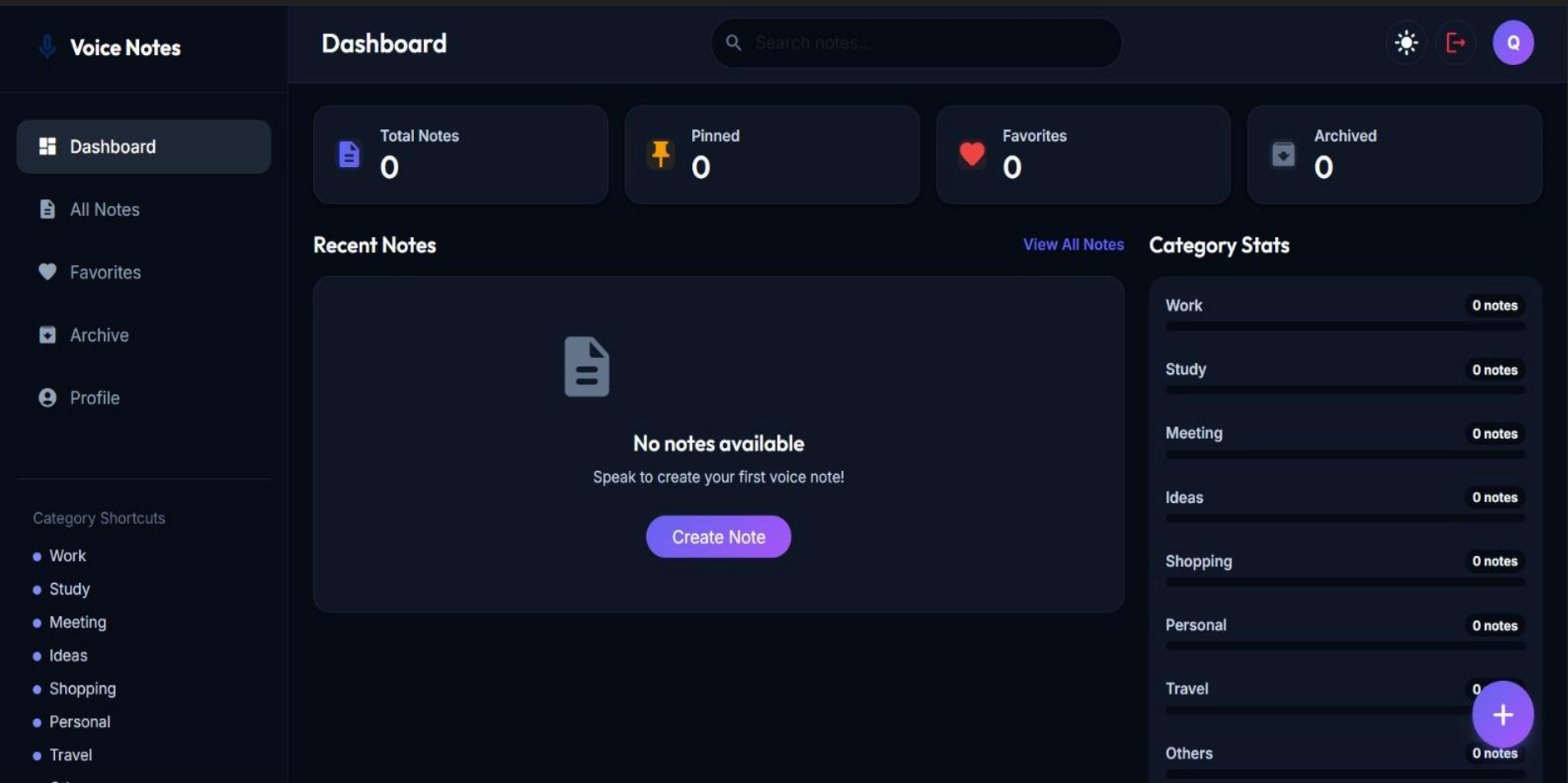
Enter 6-Digit OTP

Verify Code 

Resend code in **53s**

[← Back to Login](#)

Screenshots of Output



Screenshots of Output

 **Voice Notes**

Dashboard

All Notes

Favorites

Archive

Profile

Category Shortcuts

Work

Study

Meeting

Ideas

Shopping

Personal

Travel

Create Note

Search notes...



← Cancel

Create New Note

Enter Title...

ஜஸ்கிரீம் சிறந்த முடிவு பாலவனம்|

Category

Others

Note Color



 **Voice Transcriber**

Recognition Language

English (US)



Microphone is idle



Session Transcript Preview:

"ice cream is the best result desert"

 Listen to Draft Content

Tamil